Yaoyu Wang

Montreal, QC | Open to Fully Remote Across Canada | English (Fluent) | French (Professional) | Mandarin Chinese (Native) | +1 250-667-7900 | wyy844391606@gmail.com | linkedin.com/in/yaoyu-wang-660692232 | Canadian Permanent Resident

PROFESSIONAL SUMMARY

Senior Frontend Engineer with 7+ years of experience designing and delivering enterprise-scale, data-intensive web applications.

Specialized in React and TypeScript architecture for complex workflow systems, real-time AI-driven interfaces, and high-performance data visualization platforms in regulated environments.

Proven track record of leading frontend system design, optimizing performance for large datasets, and driving full delivery lifecycle from technical planning to production deployment in cross-functional teams.

Passionate about scalable architecture, engineering quality, and building maintainable platforms that support long-term product growth.

TECHNICAL SKILLS

Frontend: React,Next.js,TypeScript,JavaScript (ES6+), HTML5, CSS3, Redux

Architecture & Patterns:
Micro-frontend, RBAC, Design Systems, Component Reusability, Scalable Frontend Architecture, Performance Optimization

Data & Real-time:
Server-Sent Events (SSE), WebSocket, Data Visualization (SVG, Canvas), High-density Data Rendering

Tools & DevOps:
Git, CI/CD, Deployment Pipelines, Agile/Scrum

UI Ecosystem:
Ant Design, Ant Design Pro, Umi

Backend Collaboration:
RESTful APIs, Node.js (integration level)

WORK EXPERIENCE

Senior Frontend Engineer | BeOne Medicines (formerly BeiGene) | May 2024 – Present | Fulltime | Fully Remote

• Led frontend architecture design for multiple enterprise platforms across pharmaceutical research, transportation, and AI safety domains.

• Owned the full delivery lifecycle from requirement analysis and technical planning to production deployment and post-release evolution.

• Defined scalable component patterns, permission models (RBAC), and workflow-driven UI architecture in collaboration with backend and product teams.

• Provided technical guidance and system-level solutions for subsequent frontend development.

Key Achievements:
• Reduced memory usage from 4000MB to 1100MB for high-density data tables, enabling stable rendering of complex enterprise datasets.

• Built Excel-like interaction engine supporting 100+ fields per record with deterministic rendering and real-time recalculation.

• Designed and implemented a 30+ role RBAC system for AI safety workflow platforms.

• Developed real-time AI interfaces using SSE for continuous data streaming and analysis visualization.

Frontend Engineer | Orispark Software Inc. | Apr 2023 – May 2024 | Full-time | Remote

• Served as project client-side architect, responsible for system design based on URS and coordination with PDM and backend teams.

• Projects originated from various government departments and research institutions.

• Followed full delivery workflow including high-level requirement alignment, UI design, URS documentation, timeline discussion, development execution, and client acceptance.

• Delivered an internal backend management system for Tsinghua University Research Institute (government-affiliated), involving three technical modules.

• Completed development of a company self-developed invoice extraction and management system.

Full Stack Engineer | Waybo | Mar 2022 – Mar 2023 | Full-time | Onsite

- Served as project client-side full stack engineer, responsible for approximately 80% frontend and 20% backend development, as well as deployment tasks.
- Developed the company's self-developed robotic in-vehicle remote operation system, enabling real-time robot status monitoring and remote command execution.
- Worked in an agile startup environment, collaborating directly with the founder and robotics development team to align on evolving requirements.
- Delivered testing and production releases within defined milestone timelines.
- Independently handled CI/CD pipeline setup and environment deployment throughout the project lifecycle.

Frontend Developer | Lowkler | 2019 – 2022 | Full-time | Montreal

- Participated in frontend development of an e-commerce platform similar to large-scale online retail systems.
- Implemented product listing, order management, and user interaction modules.
- Collaborated with backend developers and designers to deliver feature updates and system improvements.
- Supported testing and production deployment across iterative releases.

PROJECT EXPERIENCE

Master Production Tracker | BeOne Medicines | Jun 2025 – Jan 2026

Tracking application developed for medical resource transportation teams to monitor and manage each transportation record status.

- Implemented Excel-type data interaction with extensive browser event handling including click, blur, focus, and callback condition logic.
- Each record contained 100+ fields with automatic calculation logic implemented directly on the frontend due to workflow allocation.
- Handled large-scale and highly complex user interaction and rendering condition logic.
- Built initialization system with user configuration loading including filters, column ordering, fixed column settings, and persisted view state restoration.
- Implemented overdue and data entry input automatic recalculation and state synchronization logic.

- Developed last-edit auto-positioning functionality for improved operational continuity.
- Designed row-level recalculation mechanism after backend interaction while ensuring singlecell re-rendering without affecting other data or full-table interaction.
- Applied extensive chained macrotask scheduling to guarantee ordered and deterministic execution of user operations and rendering cycles.
- Leveraged Observer-based viewport rendering control to manage cell rendering visibility.
- Reduced heap memory usage from approximately 4000MB to 1100MB under 50-rows-perpage conditions.
- Designed and implemented a role-based permission management system supporting 30+ distinct roles with fine-grained access control.
- Optimized performance for multi-row scenarios with 100-field density per record.
- Independently completed the entire frontend implementation.

Tech: Umi Max, Ant Design Pro, Ant Design, React, TypeScript, Observer API, Macro Task Scheduling, Custom Calculation Engine.

ASR (AI Safety Review System) | BeOne Medicines | Sep 2025 – Jan 2026

AI-driven safety analysis system designed to analyze existing medical documentation chapters and generate AI-assisted safety reports.

- Independently led frontend development, collaborating with Product Managers and coordinating with backend and AI engineers to define data models and API structures.
- Designed and implemented role-based workflow including Writer, Contributor, and Admin roles with structured approval processes.
- Built contributor assignment, submission, and approval workflow before AI processing.
- Developed AI report generation interface with chart-based interaction and version control management.
- Implemented comprehensive permission validation, access control (RBAC), and full audit trail tracking.
- Successfully delivered and passed internal departmental acceptance.

Tech: Umi Max, Ant Design Pro, Ant Design, React, TypeScript, RBAC, Server-Sent Events (SSE), sessionStorage (token), Umi Model (global state), microtask/macrotask controlled rendering, GitHub Copilot.

Safety Science AI | BeOne Medicines | Sep 2025 – Jan 2026

AI-based safety data analysis module designed to extract internally stored safety datasets by structured chapter segmentation and generate AI-assisted analytical reports.

- Independently implemented frontend modules aligned with the ASR system architecture.
- Developed structured data extraction and transformation logic to prepare internal safety datasets for AI processing.
- Built AI result visualization interfaces to present structured analysis outputs.
- Integrated AI report generation workflow without role-based approval structure.
- Delivered as a sub-module within the broader ASR ecosystem.

Tech: Umi Max, Ant Design Pro, Ant Design, React, TypeScript, Server-Sent Events (SSE), AI Integration.

Feasibility (Patient Demand Prediction Platform) V1 & V2 | BeOne Medicines | May 2024 – Jun 2025

Data analytics platform providing clinical teams with predictive patient demand insights by country and major city based on internal datasets.

- Independently implemented the client-side frontend aligned with enterprise system architecture.
- Built dynamic comparative data visualization for multi-region demand analysis.
- Established long-lived connection to continuously retrieve latest datasets from the server and render incremental updates in real time.
- Developed high-precision charts using native SVG and Canvas instead of third-party libraries due to internal design requirements.
- Implemented responsive chart scaling logic based on current browser width.
- Designed component rendering strategies for continuously updating analytical data.

V2 Enhancements

- Integrated shared form and functional modules from another internal system.
- Implemented micro-frontend integration for cross-project collaboration.

- Enabled data exchange via URL parameter passing and parent–child communication within the micro-frontend architecture.

Tech: React, TypeScript, Native SVG, Canvas, Long-lived Connection, Micro-Frontend Architecture.

E-Invoice System | Orispark Software Inc. | Dec 2023 – Apr 2024

Invoice management system rendering structured invoice data on the client based on backendparsed JSON.

• Independently led frontend development for invoice visualization and interaction.

• Implemented file upload and export functionality.

• Built reusable data table and form modules with filtering and sorting capabilities.

Tech: React, TypeScript, Ant Design.

HR Management System | Tsinghua Wuxi Research Institute | Mar 2023 – Dec 2023

Internal personnel management system supporting daily workforce operations.

- Developed general form-based workflows for timesheet submission, attendance tracking, and leave management.

- Implemented reusable data entry and management interfaces for internal administrative processes.

Tech: React, TypeScript, Ant Design.

HMI Robot Management System | Waybo | Mar 2022 – Oct 2022

Internal web-based management system providing remote operation interfaces for KUKA and CyberDrawer robots based on device type.

• Rendered dynamic user control interfaces according to different robot models.

• Implemented real-time robot data monitoring and remote operation capabilities.

• Built order completion tracking functionality for operational workflow visibility.

• Integrated alert mechanism for abnormal operation conditions.

Tech: React, Node.js.

Remote HMI Device Management System | Waybo | Oct 2022 – Mar 2023

Web-based system for tracking deployed HMI devices with remote version release and real-time status monitoring.

- Implemented core frontend modules for device tracking and operational control.
- Supported remote automated version deployment and live device status visualization.

Tech: React, Node.js.

Online Shopping Application | Lowkler | 2020 – 2022

E-commerce platform similar to Amazon for online purchasing and shipment tracking.

- Implemented user-facing shopping workflows and logistics tracking features.
- Built management-side interfaces for supply channel status monitoring and online customer service chat.

Tech: React, JavaScript.

EDUCATION

John Abbott College — Bachelor's Degree in Computer Science | 2014 – 2019